LAB 7 – JOINS

Create Tables and Insert

```
CREATE TABLE
Department (
dept_id INT
PRIMARY KEY,
dept_name
VARCHAR(30) );

CREATE TABLE
Employee (
emp_id INT
PRIMARY KEY,
emp_name
VARCHAR(30),
dept_id INT );

CREATE TABLE
Staff (
emp_id INT
PRIMARY KEY,
emp_name
VARCHAR(30),
manager_id INT
);

INSERT INTO Department VALUES
(10, 'Sales'), (20, 'HR'), (30, 'IT'), (40, 'Finance');

INSERT INTO Employee VALUES
(201, 'Likith Kumar', 10),
(202, 'Maanas J', 20),
(203, 'Hemanth Gowda', 30),
(204, 'Madhan Kumar', NULL),
(205, 'Aditya C', 10),
(206, 'Dhavan Ganesh', 20);

INSERT INTO Staff VALUES
(201, 'Likith Kumar', NULL),
(202, 'Maanas J', 201),
(203, 'Hemanth Gowda', 201),
(204, 'Madhan Kumar', 202),
(205, 'Aditya C', 202),
(206, 'Dhavan Ganesh', 203);
```

Inner Join

```
SELECT e.emp_id, e.emp_name, d.dept_name
FROM Employee e
INNER JOIN Department d ON e.dept_id = d.dept_id;
```

emp_id	emp_name	dept_name
201	Likith Kumar	Sales
202	Maanas J	HR
203	Hemanth Gowda	IT
205	Aditya C	Sales
206	Dhavan Ganesh	HR

Left Join and Right Join

```
SELECT e.emp_id, e.emp_name, d.dept_name
FROM Employee e
LEFT JOIN Department d ON e.dept_id = d.dept_id;
```

```
SELECT e.emp_id, e.emp_name, d.dept_name
FROM Employee e
RIGHT JOIN Department d ON e.dept_id = d.dept_id;
```

emp_id	emp_name	dept_name
201	Likith Kumar	Sales
202	Maanas J	HR
203	Hemanth Gowda	IT
204	Madhan Kumar	NULL
205	Aditya C	Sales
206	Dhavan Ganesh	HR

emp_id	emp_name	dept_name
201	Likith Kumar	Sales
202	Maanas J	HR
205	Aditya C	Sales
206	Dhavan Ganesh	HR
203	Hemanth Gowda	IT
NULL	NULL	Finance

Full Outer Join (using UNION)

```
SELECT e.emp_id, e.emp_name, d.dept_name
FROM Employee e
LEFT JOIN Department d ON e.dept_id = d.dept_id
UNION
SELECT e.emp_id, e.emp_name, d.dept_name
FROM Employee e
RIGHT JOIN Department d ON e.dept_id = d.dept_id;
```

emp_id	emp_name	dept_name
201	Likith Kumar	Sales
202	Maanas J	HR
203	Hemanth Gowda	IT
204	Madhan Kumar	NULL
205	Aditya C	Sales
206	Dhavan Ganesh	HR
NULL	NULL	Finance

Cross Join

```
SELECT e.emp_name, d.dept_name
FROM Employee e
CROSS JOIN Department d;
```

emp_name	dept_name
Likith Kumar	Sales
Likith Kumar	HR
Likith Kumar	IT
Likith Kumar	Finance
Maanas J	Sales
Maanas J	HR
Maanas J	IT
Maanas J	Finance
Hemanth Gowda	Sales
Hemanth Gowda	HR
Hemanth Gowda	IT
Hemanth Gowda	Finance
Madhan Kumar	Sales
Madhan Kumar	HR
Madhan Kumar	IT
Madhan Kumar	Finance
Aditya C	Sales
Aditya C	HR
Aditya C	IT

Aditya C	Finance
Dhavan Ganesh	Sales
Dhavan Ganesh	HR
Dhavan Ganesh	IT
emp_name	**dept_name**
Dhavan Ganesh	Finance

Self-Join

```
SELECT e.emp_name AS employee, m.emp_name AS manager
FROM Staff e
INNER JOIN Staff m ON e.manager_id = m.emp_id;
```

employee	manager
Maanas J	Likith Kumar
Hemanth Gowda	Likith Kumar
Madhan Kumar	Maanas J
Aditya C	Maanas J
Dhavan Ganesh	Hemanth Gowda